

Scaling ML-KEM to (much) higher security

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1 Introduction

After a multi-year, multi-round public competition, NIST published its first post-quantum standards FIPS-203 [18], FIPS-204 [17], and FIPS-205 [19] in 2024, specifying ML-KEM, ML-DSA, and SLH-DSA respectively. Interestingly, four of the schemes in the PQC Round 3 Submissions [16] — Kyber [7] (standardized as ML-KEM), Saber [10], NTRU LPRime [5], and FrodoKEM [6] — are closely related lattice KEMs, spanning a spectrum of algebraic structure, from FrodoKEM’s unstructured LWE, through Kyber’s and Saber’s cyclotomic modules, to the prime-degree number fields of NTRU Prime. This spectrum reflects different design principles, commonly understood as structure vs conservatism. ML-KEM’s module structure gives small keys and fast NTT arithmetic, but is also seen as a potential attack surface, so conservative designs discard it entirely [6]. FrodoKEM, for example, is based on a more conservative hardness assumption of plain LWE, which is why bodies such as BSI, ENISA, and NLNCSA point to FrodoKEM for long-term confidentiality [20]; it was also recently standardized by ISO [13] for this very reason.

It is useful to separate the two notions of conservatism this framing bundles together as they concern two independent (and today, entirely hypothetical) classes of attack. The first arises from the choice of hardness assumption. FrodoKEM rests on unstructured LWE to protect against a structure break, an attack that would exploit the algebraic structure of ring/module LWE, but not apply to plain LWE. The second involves the security margin a scheme carries above its target level, which protects against advances in the best known attacks today. This could include improved lattice reduction, dual attacks, or sharper cost models that would affect the concrete security of every lattice scheme at its current parameters, including FrodoKEM. Both of these hedges are distinct and not to be substituted; structured conservatism does nothing against the second class of attacks and no security margin protects against the first. On top of that, it’s impossible to predict which of the two threats is more likely.

Faced with this uncertainty, increasing the margin has been a historically standard response. For computationally secure schemes, concrete security estimate is the cost of the best attack known at the time, and such estimates evolve

as cryptanalysis matures. For instance, the recommended size of an RSA modulus climbed from 512 to 1024 to 2048 bits and beyond as factoring algorithms and available computation advanced [4]. Lattice estimates are not static either. The cost models used for estimation and the attacks have evolved over more than a decade [9,1], and the cost of the best known attacks continues to drift in both directions as improvements are proposed [12,14], scrutinized [11], and refined [8].

If the best known attacks continue to improve, how much margin is required remains open, and the useful question is not whether today’s KEMs carry enough but how much margin they can carry at all. This question arises because additional margin comes at a cost, requiring larger parameters and hence higher bandwidth and runtime, so no scheme can indefinitely withstand advances in cryptanalysis simply by scaling its parameters. For LWE-family KEMs there is a further constraint, since the parameters that determine security also control the decryption failure probability, so increasing the noise to gain a larger security margin necessarily affects correctness.

Crucially, what structure determines is the cost of buying additional margin. At matched security levels, an unstructured scheme already requires roughly an order of magnitude more bandwidth than a module scheme [6,7]. Since these communication costs are considered practical enough for conservative deployments, a module scheme can spend the same bandwidth budget on larger parameters and obtain a conservative security margin while retaining the efficiency advantages of NTT arithmetic.

Recent work has explored this high-margin regime from the perspective of unstructured LWE. KyFrog [15] increases the LWE dimension and adjusts the noise distribution to achieve substantially higher security estimates for much larger ciphertexts. This motivates the complementary question of how much security margin a module KEM can achieve within the communication budget of a conservative unstructured scheme. We answer this question with KAIBURR, a module KEM that scales ML-KEM’s module rank and introduces a generalized noise distribution to preserve correctness. We show that KAIBURR achieves large security margins within FrodoKEM’s communication cost while running substantially faster.

1.1 Our contributions

1.2 Structure of the paper

1.3 Access to artifacts

All software, including parameter analysis scripts, implementation source code, and other artifacts, is publicly available on GitHub.

2 Preliminaries

3 Warmup: Just changing k

The main mechanism to increase bit security in Kyber is to raise the module rank k [2]. Fixing the ML-KEM base ring, noise distribution, and modulus, and varying only k , we measure the decryption-failure probability using the scripts provided by KYBER team⁵. Throughout we use the centered binomial noise CBD_2 (the noise of ML-KEM-768 and -1024; recall that ML-KEM-512 uses CBD_3). The ranks $k = 3$ and 4 recover the standardized sets ML-KEM-768, and -1024, while $k \geq 5$ is an extrapolation. Table 1 reports the results. Against the 2^{-128} failure bound expected at this security level, the bound holds at $k = 7$ ($2^{-133.7}$) but fails at $k = 8$ ($2^{-116.8}$), so raising k further cannot buy higher security without sacrificing correctness.

These values are computed without ciphertext compression, and therefore are an optimistic estimate. The rounding applied to $\mathbf{u}$ and v in Kyber introduces additional error, which can only increase the true failure probability. Hence, the actual failure probabilities are expected to be higher than those reported here, and the correctness limit on k may be even more restrictive.

Table 1. Decryption-failure probability and public-key and ciphertext sizes as a function of module rank k , at fixed ML-KEM noise and modulus, without ciphertext compression. Ranks $k = 3, 4$ correspond to ML-KEM-768, and -1024.

k	Failure probability	pk (bytes)	ct (bytes)
3	$2^{-299.6}$	1184	1536
4	$2^{-231.1}$	1568	1920
5	$2^{-186.5}$	1952	2304
6	$2^{-155.9}$	2336	2688
7	$2^{-133.7}$	2720	3072
8	$2^{-116.8}$	3104	3456

4 Revisiting the noise distribution

Even though the resulting public key and ciphertext sizes at $k = 7$ remain relatively small, increasing k further is not possible without affecting correctness. Instead, we could gain additional security by increasing either the ring dimension n or the modulus q . We note that all variants of Kyber fix these parameters to $n = 256$ and $q = 3329$. The ring dimension is fixed at $n = 256$ because the scheme encapsulates a 256-bit key, so that a single coefficient carries a single key bit. A smaller n would require packing several key bits per coefficient, forcing

⁵ <https://github.com/pq-crystals/security-estimates>

lower noise and hence lower security, while increasing n would make scaling security through k less straightforward [2]. The modulus $q = 3329$ is the small prime satisfying the NTT-enabling condition $n \mid (q - 1)$, notably weaker than the $2n \mid (q - 1)$ one might expect, which Kyber deliberately does not require [3]. It was reduced from the round-1 value 7681 to recover the bandwidth spent by dropping public-key compression and co-tuned to balance ciphertext size, failure probability, and security.

This is not to say that k is the only parameter that can be modified. We focus on changes that preserve the core structure of Kyber: the ring $\mathbb{Z}_q[X]/(X^{256} + 1)$, the NTT implementation and its constants, and the existing analysis framework. The module rank k qualifies trivially. So does the error distribution. Since the noise only affects the error term and does not interact with the arithmetic structure or the NTT, it can be replaced without modifying the core implementation. What it changes are only the MLWE hardness estimate and the decryption-failure probability, both of which can be recomputed through the same scripts, over the same ring.

In this sense the error distribution is the one degree of freedom that moves the security and correctness without changing what the scheme is, which is exactly what we exploit. Changing n or q , however, would require modifying the underlying structure of Kyber, giving a different scheme altogether. This would go beyond our goal of increasing security margin while preserving Kyber’s efficiency.

Specifically, to move beyond the warmup security ceiling, we need a noise distribution that reduces decryption failures as k increases while remaining as efficient to sample as Kyber’s centered binomial distribution. In particular, it should still admit a simple branchless, constant-time sampler. This rules out simply adopting a discrete Gaussian, as in FrodoKEM.

4.1 A parameterized noise distribution

We introduce a one-parameter family $f(t)$ on $\{-2, -1, 0, 1, 2\}$, defined by

$$\Pr[\pm 2] = 2^{-t}, \quad \Pr[\pm 1] = \frac{1}{4}, \quad \Pr[0] = \frac{1}{2} - 2^{-(t-1)}.$$

The probabilities at ± 1 remain fixed, while the probability mass at ± 2 decreases exponentially with t .

Samples from $f(t)$ can be generated using Algorithm 1, which requires only uniform random bits and simple bitwise operations. The sampler is branchless and therefore runs in constant time. We machine-checked both constant-time (CT) and speculative constant-time (SCT) in Jasmin [?].

The standard deviation is

$$\sigma(t) = \sqrt{0.5 + 2^{3-t}},$$

which decreases from 1 at $t = 4$ toward $\sqrt{0.5}$ as t increases. $f(t)$ can also be seen as a generalization of Kyber’s noise distribution. In particular, $f(4) = \text{CBD}_2$,

Algorithm 1: Branchless sampler for $f(t)$

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1 Sample independent bits  $a_1, \dots, a_t \in \{0, 1\}$ ;
2  $s \leftarrow 2a_t - 1$ ;
3  $m \leftarrow 1 - a_1 + 2 \prod_{i=1}^{t-1} a_i$ ;
4 return  $s \cdot m$ ;

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the distribution used in ML-KEM-768 and ML-KEM-1024, and

$$\lim_{t \rightarrow \infty} f(t) = \text{CBD}_1.$$

4.2 Why thinning the tail is an efficient trade

The goal of $f(t)$ is not simply to reduce the overall noise, but to selectively reduce the part of the distribution that contributes most to decryption failures. In Kyber’s centered binomial distribution, the size of the noise is controlled by a single parameter η , which changes both the main mass and the tail of the distribution. In contrast, $f(t)$ keeps the probability at ± 1 fixed while reducing the heavier ± 2 tail. This lets us decrease the variance that drives failures without removing all of the noise contributing to MLWE hardness.

A further advantage of the one-parameter family is that it provides a continuous range of noise levels. Kyber’s centered binomial distributions only provide the two choices $\eta = 1$ and $\eta = 2$, with standard deviations $\sigma \approx \sqrt{0.5}$ and $\sigma = 1$, respectively. By varying t , $f(t)$ smoothly fills the range between these two extreme values. This allows the noise level to be tuned for each module rank k rather than selecting a fixed distribution that may be unnecessarily conservative.

4.3 Limit of the noise family

The flexibility provided by $f(t)$ is not unlimited. The smallest noise achievable by this family is the limiting distribution $f(\infty)$, where

$$\Pr[\pm 1] = \frac{1}{4}, \quad \Pr[0] = \frac{1}{2}, \quad \Pr[\pm 2] = 0,$$

with minimum variance $\sigma^2 = 0.5$. Even at this limit, the accumulated error eventually causes the decryption failure probability to exceed the 2^{-128} threshold. For $n = 256$ and $q = 3329$, this occurs at $k = 30$, so $k = 29$ is the largest rank supported by the entire family. Beyond this point, further increases in security would require changing other parameters such as n or q .

4.4 Selecting t

For each rank k , correctness and security define an acceptable range of values for t . We select representative values within this range based on implementation

Table 2. Failure probability under the limiting distribution $f(\infty)$ (minimum variance $\sigma^2 = 0.5$) as k increases, uncompressed. The 2^{-128} bound is first violated at $k = 30$, so $k = 29$ is the limit of the $f(t)$ family at $n = 256$, $q = 3329$.

k	Failure probability	pk (bytes)	ct (bytes)
24	$2^{-158.7}$	9248	9600
26	$2^{-146.2}$	10016	10368
28	$2^{-135.5}$	10784	11136
29	$2^{-130.6}$	11168	11520
30	$2^{-126.1}$	11552	11904

efficiency. Since t determines the number of random bits used per sample, small byte-aligned values provide a natural trade-off between sampling cost and noise reduction. Some of these values are natural: $t = 4$ recovers the CBD₂ sampler and serves as our anchor; $t = 8$ admits one-byte-per-coefficient SIMD packing; and $t = 6$ sits between them, spending fewer PRF bytes and offering more security per dimension at the cost of less regular packing. For each selected value of t , we then increase k until reaching the largest rank that satisfies the decryption-failure bound. The theoretical limit of the noise family is $k = 29$, but these larger ranks are not practical because their communication cost exceeds our target FrodoKEM-640 bandwidth (9616 bytes pk, 9720 bytes ct). As a result, we choose parameter sets that balance security, correctness, and communication: $k = 24$ for $t = 8$, with analogous choices for $t = 6$ and $t = 4$ (Table 3).

Table 3. KAIBURR parameter sets. Classical/quantum security (Core-SVP) and failure probability computed with the Kyber scripts; other parameters as in ML-KEM, compression removed.

Scheme	k	noise	failure	class.	quant.	pk (B)	ct (B)
kaiburr4	7	$f(4)$	$2^{-133.7}$	479	434	2720	3072
kaiburr6	18	$f(6)$	$2^{-134.1}$	1281	1162	6944	7296
kaiburr8	24	$f(8)$	$2^{-139.9}$	1717	1557	9248	9600

5 Benchmarks

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